Lesson plan · for the teacher

Back it up: lose a phone, not your work

Cyber security · Level 2 · Lesson 2

Time
40 min

Materials
Board and chalk; Paper and pencils for each group; A USB drive or a phone to show (optional)

Learning goals

- Explain what a backup is and why we need more than one copy.
- Use the 3-2-1 rule to plan where copies live.
- Explain why one copy should stay unplugged.

Activity: Where does the register live?: Three copies

1. Warm-up (5 min): Tell the story: a teacher's phone falls into a bucket of water at break, with three years of photos, the marks and every parent's number. Ask pupils what was lost, and what could have saved it.
2. Teach (8 min): Draw a phone, a cloud, a school and a house. Explain the 3-2-1 rule: three copies, two kinds of storage, one away from school. Add a USB drive that is plugged in only to make the copy.
3. Three copies (15 min): Each group draws where three copies of the class register could live and labels each one. Walk around and ask: what if the school floods? What if the computer is locked?
4. Share (7 min): Groups show their drawings. They circle the copy that survives a flood and the one that survives ransomware, and explain why.
5. Wrap-up (5 min): Exit ticket: each pupil writes one thing they would back up and where its copies would live.

Before the lesson, draw a phone, a cloud, a school and a house on one side of the board. Prepare paper and pencils for each group. If you have a USB drive, bring it to show.

Discussion

- What would you lose if your phone disappeared today?
- Why isn't a synced copy enough?
- Where could the school keep a copy away from the building?

Exit ticket

- One thing I would back up + where its copies would live.

Try it at school tomorrow
Tell the story of a phone that fell in a bucket of water. In groups, pupils draw three places a copy of the class register could live (the phone, the cloud, a USB drive at home), then circle the copy that would survive a flood at school and the one that would survive ransomware.

Practice questions · for you

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1 **Ms Amina keeps her marks on her phone, with a copy in the cloud. Does that meet the 3-2-1 rule?**

- ◯ No: she needs a third copy, unplugged and away from school
- ◯ Yes: two copies are enough
- ◯ Yes, as long as the cloud copy syncs every minute

2 **Put a teacher's weekly backup routine in order.**

- ☐ Copy this week's marks and plans to it
- ☐ Unplug the drive and lock it away
- ☐ Plug in the USB drive
- ☐ Open one copied file to check it

3 **If your files sync to the cloud, you don't need any other backup.**

◯ True ◯ False

4 **Your backup has little space. Which of these most need to be in it?**

- ☐ Marks and class registers
- ☐ Parents' phone numbers
- ☐ Apps you can download again from the store
- ☐ Lesson plans and exam papers you wrote
- ☐ Videos you downloaded and can find online again

5 **Match each copy to the bad day it protects you from.**

The phone's automatic cloud backup	• • • •	Ransomware on the office computer
A USB drive unplugged in the office drawer	• • • •	This morning's work saved over by mistake
A copy kept at home	• • • •	A stolen phone: setting up the new one
Version history	• • • •	A fire or flood at school

✎ **What is the one file you couldn't bear to lose? Where are its copies today, and which part of 3-2-1 is missing?**

Check your answers

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1 **No: she needs a third copy, unplugged and away from school**

Phone plus cloud is two copies. The rule asks for three, and an unplugged copy protects against mistakes and ransomware that a syncing copy would simply copy.

2 **Top to bottom: 2, 4, 1, 3**

Copy, check, then unplug. A copy you haven't opened may be incomplete, and a drive left plugged in can be locked by ransomware along with everything else.

3 **False**

Sync copies mistakes too: a file deleted or locked on your phone is deleted or locked in the cloud.

4 **Marks and class registers; Lesson plans and exam papers you wrote; Parents' phone numbers**

Back up what you can't get again: marks, registers, your own plans and exam papers, and contacts.

5 **The phone's automatic cloud backup — A stolen phone: setting up the new one**
A USB drive unplugged in the office drawer — Ransomware on the office computer
A copy kept at home — A fire or flood at school
Version history — This morning's work saved over by mistake

Each copy covers a different bad day: the phone's backup rebuilds a new phone, the unplugged drive is out of ransomware's reach, the copy at home survives a disaster at school, and version history brings back an earlier version of the file.